

Chapter 4 — An End-to-End Empirical Study of Solana ↔ Ethereum Cross-Chain Arbitrage

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1 LANDSCAPE OF SOLANA ↔ ETHEREUM BRIDGES

Five general-purpose bridges currently support sustained asset transfers between the Solana and Ethereum ecosystems: Wormhole Portal, LayerZero V2, Mayan Finance, Relay, and Chainlink CCIP. They differ substantively in trust model, fee schedule, and verification mechanism, resulting in distinct latency, fee, and per-transfer-cap profiles. Table 1 summarises their main parameters alongside the volume of bridge events we were able to collect for each.

Table 1: Comparison of SOL ↔ ETH bridges

Protocol	Trust / Verification	Fee model	Records
Wormhole Portal	Guardian-signed VAA	Gas-based	42,181
LayerZero V2	DVN verification	DVN fees	271,813
Mayan Finance	Dutch auction	~3 bps	10,653,600
Relay	Intent-based fill	Variable bps	107,220
Chainlink CCIP	DON + risk manager	LINK fees	2,115

Among these, Wormhole and CCIP adopt an attester-based “messaging layer + token wrapping” architecture; LayerZero is layered across DVN attestation and Oracle/Relayer verification; Mayan and Relay model cross-chain settlement as an “intent/auction” in which solvers compete to fill orders. These architectural differences determine: (i) whether cross-chain events carry on-chain, 1:1-verifiable cryptographic fingerprints; (ii) whether each transfer can be paired into an (src, dst) tuple through a single public field; and (iii) whether there is intermediary-priced latency-fee heterogeneity. These three properties are exactly the preconditions for reconstructing per-candidate arbitrage PnL in the remainder of the chapter.

2 WHY WORMHOLE: DATA AND CANDIDATE CONSTRUCTION

2.1 Why Wormhole Portal

Of the five bridges above, we use Wormhole Portal as our sole long-horizon sample source. Our reasoning is fourfold: (i) *On-chain verifiability*. Every Portal message is cryptographically signed by the Guardian cluster and committed on-chain as a Verifiable Action Approval (VAA),

so that a Solana signature and an Ethereum transaction hash can be paired 1:1 from cryptographic evidence alone, without any off-chain ledger; (ii) *Standardised pairing schema*. The Token Bridge contracts on both chains use fixed emitter/recipient field layouts, so the “actor” primary key (arb_{sol}, arb_{eth}) can be decoded directly from on-chain data; (iii) *Decoupled transport vs. execution*. Portal handles only token wrap/unwrap; any subsequent DEX execution on the destination chain is an independent on-chain event, allowing entry-swap, bridge-leg, and exit-swap to be attributed separately; (iv) *Widest token coverage*. For the long-tail memecoins that dominate observed cross-chain arbitrage, the token set supported by Wormhole (via its WTT and NTT standards) is a strict superset of the other four bridges, so “Wormhole-only” sampling introduces no material selection bias for the phenomenon we study.

2.2 Data Collection Pipeline

Figure 1 shows our five-stage collection and matching pipeline. The official Wormhole /operations API indexes by VAA but *only retains approximately 7 days of history*, which is insufficient for our year-long analysis. We therefore *invert the pipeline* and start from the on-chain Portal contracts on both legs: (1) on Ethereum and Solana, scan the Portal Token Bridge contract’s full log/instruction stream and parse the `src_chain_id` and `dst_chain_id` of every bridge event; (2) for each event, use the Solana signature or Ethereum transaction hash as a query key against Wormholescan’s sig/hash search endpoint to retrieve the corresponding VAA (this path is not bounded by the 7-day window of the /operations API); (3) merge the two-chain signatures/hashes into a unified cross-chain record keyed by the VAA identifier (`emitterChain + sequence`); (4) query Birdeye’s 1-second price endpoint for the token-USD rate at the entry and exit timestamps; (5) run a greedy attribution algorithm to assign same-wallet swaps immediately before/after the bridge event to the candidate, yielding tuples of the form (direction, sol_sig, eth_hash, entry_swaps, exit_swaps, C_{exec}^{in} , Π_{net}).

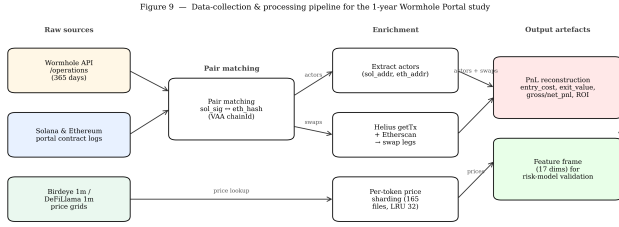


Figure 1: Data collection and arbitrage-candidate construction pipeline.

3 WHAT DRIVES CROSS-CHAIN PROFIT: DESCRIPTIVE EVIDENCE

This section answers the question: descriptively, which observables are correlated with *net PnL* Π_{net} and which are not? Figure 3 reports six sub-results at the economic level.

Distribution shape (panels a/b). ROI is heavy-tailed and left-skewed, with median +1.39% and 78.0% of samples positive; the SOL $\rightarrow$ ETH CDF has a slightly thinner left tail and ETH $\rightarrow$ SOL a slightly thicker one, but the two directional medians differ by less than 1 percentage point, so direction itself has limited explanatory power. Table 2 reports the full quantile structure for reliable candidates. The heavy-tailed profile is unambiguous: p_1 sits at -49% while p_{99} reaches +152%; the ratio $\text{std}/|\text{median}|$ exceeds 48; the mean +7.83% is $5.6\times$ the median +1.39%, dragged by a small number of extreme positive samples.

Table 2: ROI quantiles on reliable candidates ($n = 29,340$, units %)

Stat.	Value	Quantile	Value	Quantile	Value
count	29,340	p_1	-49.65	p_{75}	+3.77
mean	+7.83	p_5	-4.15	p_{90}	+10.66
std	67.07	p_{10}	-1.51	p_{95}	+27.79
min	-125.28	p_{25}	+0.20	p_{99}	+152.40
max	+5,204.09	p_{50}	+1.39	—	—

The minimum ROI of -125.28% (loss exceeding 100%) is not a calculation error: when $f_{\text{total}} > C_{\text{exec}}^{\text{in}}$, the ratio $\text{ROI} = \Pi_{\text{net}}/C_{\text{exec}}^{\text{in}}$ can mathematically cross -100%. The most extreme LOVE sample has entry \$1.00 and fees \$1.13—a genuine economic loss, not a measurement artefact.

Capital is essentially unrelated to PnL (panels c/f). On a logarithmic x-axis, entry capital spans seven decades. Fitting an OLS regression

$$\Pi_{\text{net}} = \alpha + \beta \cdot \log_{10}(C_{\text{exec}}^{\text{in}}) + \varepsilon \quad (1)$$

on the $n = 29,128$ reliable candidates yields $\hat{\beta} = -1,085$ \$/decade with $R^2 = 0.0027$. In words, *entry capital explains only 0.27% of the variance in net PnL*. The binned-median curve in panel (f) is approximately flat and slightly positive over the 10^2 – 10^4 USD band and sinks at both ends—an inverted-U shape. This counter-intuitive result refutes the folk assumption that cross-chain arbitrage is a linearly capital-scalable business.

Fees are a first-order constraint (panels d/e). The fee-rate distribution is extremely wide; the median $\text{fee}/|\text{gross}| = 0.053$, but 2.8% of candidates suffer $\text{fee} > |\text{gross}|$ —severe fee erosion. In panel (e) every point lies strictly below the $y = x$ line; the vertical displacement visualises the fee cost directly, and the dense lower-left cluster corresponds to the many small-volume losses whose gross margin failed to cover gas.

Temporal structure (Fig. 2). Figure 2 characterises the time-side properties of cross-chain candidates from three angles: (a) bridge-latency CDF by direction—SOL $\rightarrow$ ETH (blue) has median ~ 34 s while ETH $\rightarrow$ SOL (red) has median $\sim 1,011$ s (~ 17 min), a $\sim 30\times$ asymmetry that matches the gap between Solana’s sub-second slot finality and Ethereum’s two-epoch finality window; the two CDFs converge only in the far right tail (both $p_{99} > 4,000$ s); (b) swap-to-bridge time gap—the entry-side gap has a median ~ 12 s while the exit-side is longer (~ 24 s), with a right tail extending into the hour range (a small number of candidates hold the bridged position before exit), which justifies attributing pre/post bridge swaps of the same wallet to the same cross-chain arbitrage candidate; (c) ROI boxplot by latency tier—medians are non-monotone across tiers: < 60 s ($n = 14,599$, median +1.39%) and 60–300 s ($n = 2,639$, median +1.04%) both outperform the wider 300 s–1 h tier ($n = 14,449$, median +1.50%), while the 1–24 h tier collapses to +0.13% and the > 24 h tier to -1.35%. Panel (c) is the direct univariate evidence for the “bridge latency negative-significant” row of Table 3.

Summary of “what matters / what does not.” Table 3 collects the descriptive evidence above. These are all *univariate* claims; some of the variables are re-evaluated under the multivariate model in Section 4.

4 INTEGRATED MODELLING AND EXPERIMENTS

The conclusions in §3 are all univariate. If the features interact, univariate relations can mislead. This section reassesses both the direction and magnitude of Π_{net} via a multivariate non-linear model under two parallel targets: a *regression* predicting the dollar amount of Π_{net} , and a *classification* predicting $1\{\Pi_{\text{net}} > 0\}$.

Figure 2 — Timing of Wormhole Portal Arbitrage

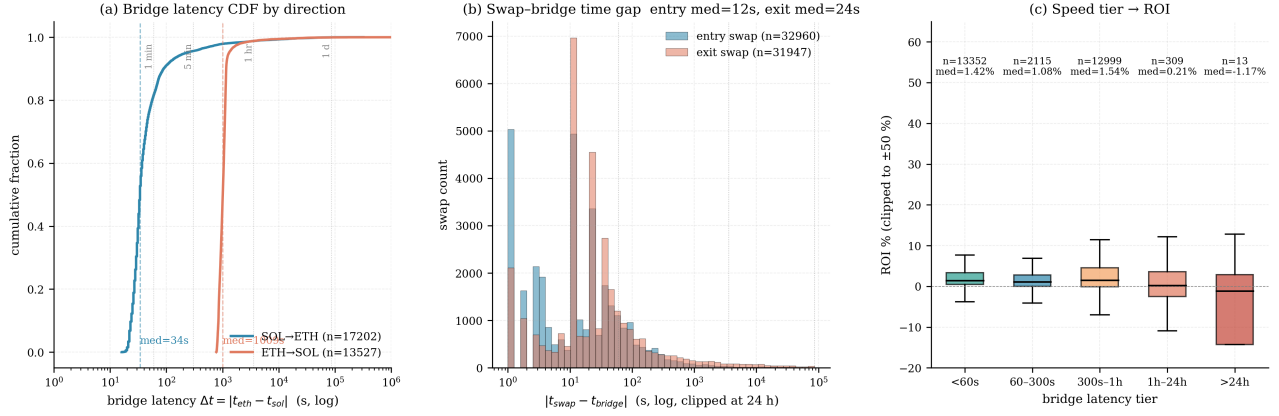


Figure 2: Temporal structure of Wormhole Portal arbitrage: (a) bridge-latency CDF by direction; (b) swap-bridge time gap for entry and exit swaps; (c) ROI distribution stratified by bridge-latency tier.

Figure 3 — Economics of Wormhole Portal Arbitrage (ROI, Size, Fees, PnL)

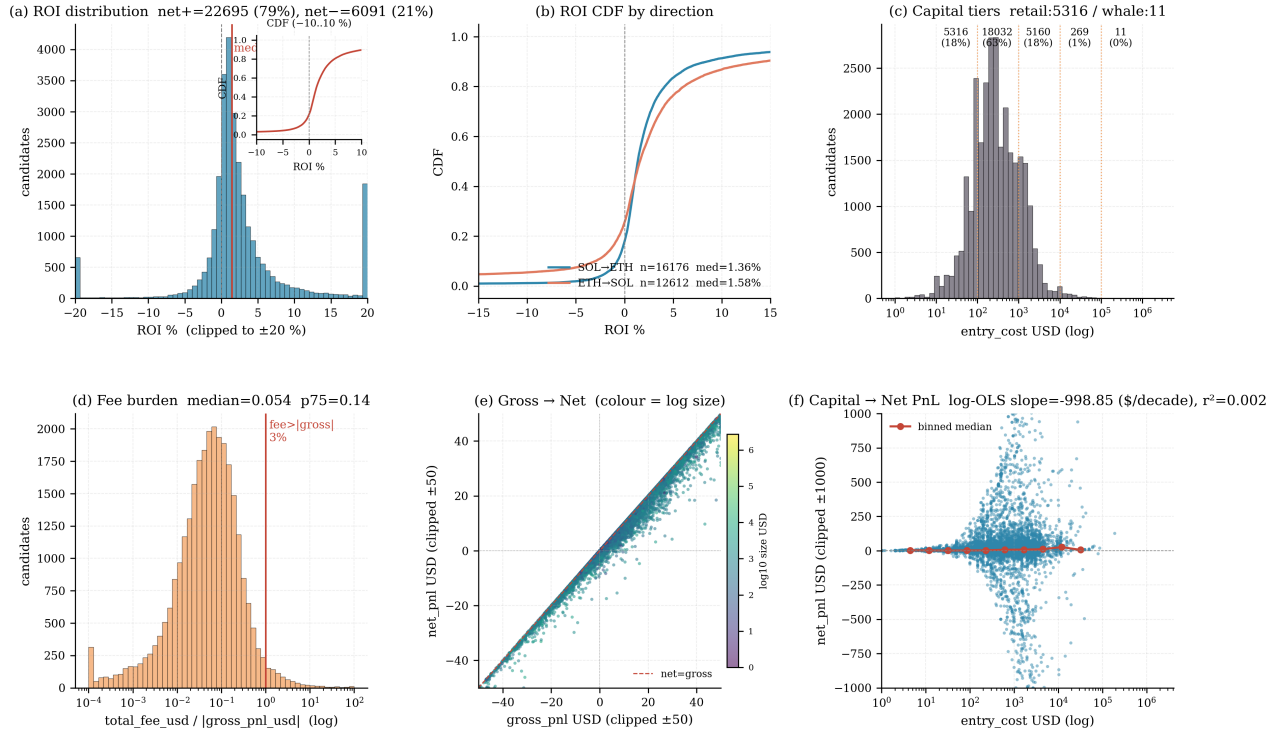


Figure 3: Economics of cross-chain arbitrage: (a) ROI histogram with inset CDF; (b) ROI CDF by direction; (c) log-scale distribution of entry capital and tier shading; (d) distribution of $fee/|gross\ PnL|$; (e) Gross vs. Net PnL scatter (colour-encoded by capital); (f) OLS regression of Π_{net} on $\log_{10}(C_{exec}^{in})$.

4.1 Feature set

We assemble 17 features covering six hypothesised driver families, all observable at the candidate level: *direction / time*

Table 3: Descriptive relations between Π_{net} and each variable

Variable	Relation	Evidence
Entry spread	Positive, significant	Prior work + tiering
Bridge latency	Negative, significant	Fig. 2(c) + model
Liquidity depth	Non-linear	Liquidity-tier strata
Fees	Negative (identity)	Fig. 3(d/e)
Entry capital	$R^2 = 0.003$, near-null	Fig. 3(f)
Direction	Gap < 1 pp	Fig. 3(b)
Hour of day	Approximately uniform	(omitted)
Swap split count	No significant effect	Fig. 2(b)

(dir_sol_to_eth, hour_utc); size / latency (log_size, log_latency); atomicity (atomic_entry_same_block, atomic_exit_same_block); swap-bridge timing (entry_delta_bridge_sec, exit_delta_bridge_sec, n_entry_swaps, n_exit_swaps); measurement completeness (entry_coverage, exit_coverage); liquidity tiers and market (liq_dead, liq_thin, liq_healthy, entry_gap_pct, actor_trade_count).

4.2 Model specification

Both models share the same additive tree ensemble:

$$F_M(x) = \sum_{m=1}^M \eta \cdot h_m(x), \quad (2)$$

where h_m is a depth-3 regression tree, $M = 200$, and $\eta = 0.05$. Each tree fits the negative functional gradient of the current loss:

$$h_m = \arg \min_h \sum_{i=1}^n L(y_i, F_{m-1}(x_i) + h(x_i)), \quad (3)$$

with subsample = 0.8 applied per tree to reduce variance.

Regression target (“how much”). The target is a USD-winsorised Π_{net} :

$$y_i^{\text{reg}} = \text{clip}(\Pi_{\text{net},i}, -\$100, +\$500), \quad (4)$$

with squared-error loss $L_{\text{reg}}(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$. The bounds $[-\$100, +\$500]$ are chosen for four reasons: (i) the raw Π_{net} distribution is extremely heavy-tailed (min $\approx -\$2.07\text{M}$, max $\approx +\$428\text{K}$), so without clipping a single outlier contributes $O(10^{12})$ gradient under MSE and dominates the fit; (ii) $[-\$100, +\$500]$ contains 95.8% of candidates (only 2.5% below, 1.7% above), i.e. almost all samples keep their original values and only the tails are winsorised; (iii) the bounds are deliberately asymmetric because the distribution itself is right-skewed with median $+\$3.14$ and p75 $\approx +\$10$, while the losing tail concentrates around the per-trade fee scale ($\sim \$20-50$), so $-\$100$ already corresponds to “clearly bad” and $+\$500$ preserves the heterogeneity of winners up to the

top $\sim 3\%$; (iv) the clip applies only to the regression head; the classification target y^{cls} still sees raw signs, so no information about large losses or large gains is lost at the system level.

Classification target (“whether profitable”). The target is a profit indicator:

$$y_i^{\text{cls}} = \mathbf{1}\{\Pi_{\text{net},i} > 0\}, \quad (5)$$

with binary cross-entropy $L_{\text{cls}}(y, \hat{p}) = -y \log \hat{p} - (1 - y) \log(1 - \hat{p})$, producing a probability $\hat{p}_i = \sigma(F_M(x_i))$. A trading decision is obtained via a threshold τ : $\hat{y}_i^{\text{cls}} = \mathbf{1}\{\hat{p}_i > \tau\}$.

4.3 Evaluation protocol

The training set is restricted to candidates with pnl_complete and no missing features, giving $n = 15,497$. We use $K = 5$ -fold shuffled KFold (random_state= 42): each fold trains on 4/5 of the data and tests on the held-out 1/5. Cross-validated scores are averaged across folds:

$$R_{\text{CV}}^2 = \frac{1}{K} \sum_{k=1}^K R_k^2, \quad \text{AUC}_{\text{CV}} = \frac{1}{K} \sum_{k=1}^K \text{AUC}_k. \quad (6)$$

Regression reports R^2 , MAE, Median AE; classification reports AUC, log-loss, and at two operating thresholds $\tau \in \{0.5, 0.8\}$,

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}. \quad (7)$$

The naive classification baseline is “always predict profit,” whose accuracy equals the positive rate $\bar{y}^{\text{cls}} = 73.85\%$.

4.4 Cross-validated results

Table 4 summarises 5-fold CV performance. The regressor reaches $R_{\text{CV}}^2 = 0.799 \pm 0.027$ —more than two orders of magnitude above Eq. (1)’s univariate $R^2 = 0.003$, proving that Π_{net} is predictable given the 17-feature set, but only under a multivariate non-linear model. Median AE = $\$5.13$ implies at least half of the samples are predicted within $\pm \$5$; the gap between MAE = $\$17.14$ and Median AE (a factor of 3.3) indicates a fat-tailed error distribution—a minority of extreme samples remain under-fit, confirming the need for the clipping in Eq. (4).

The classifier’s AUC is 0.866 ± 0.005 : substantially above the 0.5 random baseline but well below the 1.0 perfectly-separable limit—reflecting intrinsic randomness in cross-chain arbitrage (counterparty flow, Guardian-signing jitter) that the present feature set cannot exhaust. At $\tau = 0.5$, accuracy is 82.3%, +8.5 pp over the naive $\bar{y}^{\text{cls}} = 73.85\%$ baseline. Raising τ to 0.8 trades recall for precision: precision rises to 93.5% while recall falls to 66.2%—only trading when the model is > 0.8 confident yields a $\sim 94\%$ realised-win rate at the cost of missing 34% of the opportunities. This precision-recall schedule directly motivates using the classifier as a

high-confidence strategy-side filter. Log-loss = 0.386 is also below the binomial baseline of 0.576.

Table 4: Model performance (5-fold CV, $n = 15,497$)

Metric	Value
Regressor R_{CV}^2	0.7987 ± 0.0269
Regressor MAE	\$17.14
Regressor Median AE	\$5.13
Classifier AUC_{CV}	0.8658 ± 0.0052
Classifier accuracy at $\tau = 0.5$	82.3% (baseline 73.9%)
Classifier precision at $\tau = 0.8$	93.5% (recall 66.2%)
Classifier log-loss	0.3864

5 LIMITATIONS

Sensitivity to the dust-filter threshold. We set the dust-filter threshold at $C_{exec}^{in} < \$1$, motivated by the \$1–\$5 physical floor of Ethereum L1 swap gas. Raising the threshold to \$5 increases the set of excluded rows from 12 to 116 ($\sim 0.4\%$), lowers the ROI mean on the reliable subset from +7.83% to +6.88%, while leaving the median essentially unchanged ($+1.39\% \rightarrow +1.38\%$). The GBM results of Section 4 (R^2 , AUC, Median AE) do not change materially under either threshold. We keep the more conservative \$1 threshold to maximise sample size and avoid discarding legitimate small-size trades where “fees exceed entry” is a real economic loss.

Single-bridge sample bias. We collect candidates only from Wormhole Portal, leaving LayerZero V2, Mayan, Relay, and CCIP out of scope. Section 1 argued that Wormhole has the broadest memecoin coverage and the largest candidate volume among the five, and that its “messaging layer + token wrapping” structure permits exact PnL reconstruction. That said, Portal’s end-to-end latency is a median of ~ 13 s (Solana finality) plus ~ 13 min (two Ethereum epochs)—numbers specific to Wormhole. On LayerZero or intent-based Mayan/Relay, the latency mechanism, tail shape, and the arbitrage window differ substantially, so the chapter’s model coefficients and R_{CV}^2 should not be extrapolated to other bridges; a rigorous cross-bridge comparison is future work.

Observation window and regime shifts. The dataset spans approximately 365 days, over which Solana/Ethereum fee regimes, Wormhole’s Guardian signing cadence, and the behaviour of top bots may all have undergone regime shifts. The model in this chapter is a *mixed-period* average and is not segmented; splitting the window into distinct regimes may cause both model coefficients and CV scores to drift. Regime identification and segment-wise modelling are left to future work.